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Cyber Threat Detection based on Artificial Neural Networks using Event Profiles

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Abstract*—In the new age of increasingly growing cyber threats, the rule-based security measures can no longer help identify complex attacks. The given project introduces a smart cyber threat detection system using the Artificial Neural Networks (ANNs) and structural profiling of events by identifying anomalous activities in network settings. The proposed system is capable of reading security event logs and converting them to small event profiles and subsequent analysis of large-scale data by a deep learning model in order to identify patterns. Implementing the Bayesian inference into the system, it models interdependencies of variables with the help of the directed acyclic graph, which enhance the probabilistic threat classification. This hybrid solution overcomes the drawbacks of other available solutions since it improves the accuracy of detection, false positives, and provides the opportunity to analyze in real time. It uses Python, Django, and MySQL as the implementation of such a system and a number of machine learning algorithms such as ANN, CNN, Random Forest, and SVM to validate it. The experimental findings also prove the strength of the detection process of neural networks in spotting complex cyber threats providing an extensible and anticipatory approach to the current cybersecurity issues.

Keywords:Cybersecurity, Artificial Neural Network, Event Profiling, Intrusion Detection, Machine Learning, Bayesian Inference, Deep Learning, Threat Detection, Network Security, Anomaly Detection.

I. INTRODUCTION

The rapid expansion of the internet, cloud computing, and connected devices has significantly elevated the scale and complexity of cybersecurity threats. Organizations today face an unprecedented volume of cyberattacks ranging from phishing, malware, and ransomware to advanced persistent threats (APT). Conventional security mechanisms—primarily rule-based intrusion detection systems—struggle to detect sophisticated, evasive, and zero-day attacks due to their limited adaptability and dependency on predefined signatures.

Artificial intelligence (AI), especially, machine learning (ML) and deep learning (DL) algorithms have proven to be a useful method of addressing a weakness of the conventional cybersecurity systems. Among them, artificial neural networks (ANNs) have come into the picture because of having the ability to describe complex non-linear relationships as well as learning within large-scale data. Because the detection of malicious behavior is based on subtle and sometimes unconditional behavior patterns, ANNs can be trained to look out for the previously unknown patterns.

In this research, a novel approach is proposed that combines ANN models with structured event profiles to detect cyber threats in real time. Event profiling involves transforming raw security logs into concise, feature-rich representations of system events. These profiles allow for efficient input to neural network models, minimizing noise and dimensional complexity while retaining the essential behavioral features required for accurate threat detection.

To enhance the decision-making capability of the system, the architecture incorporates Bayesian inference. This probabilistic modeling approach enables the system to visualize dependencies between variables using directed acyclic graphs, thus improving classification and interpretability. Unlike black-box models that lack transparency, the integration of Bayesian networks provides an additional layer of reasoning for security analysts.

The given system works with large amounts of event data that are produced by components of security infrastructure including Intrusion Prevention Systems (IPS), firewalls, and Security Information and Event Management (SIEM) tool. The ANN-based model preprocesses these events, profiles them, and analyses them in a bid to identify the existing and unseen threats. The system produces a threat rating together with confidence measures, which can be used to facilitate quick defense.

To validate the effectiveness of the system, a number of classifiers are trained on and evaluated on real world datasets, i.e., Random Forest (RF), Support Vector Machine (SVM), Convolutional Neural Networks (CNN) and ANN. ANN-based method has been found to be more precise, contains fewer false positive results and is able to accommodate increased degree of flexibility to new pattern of attacks. Comparative results show that ANN along with event profiling scaled well and what is more, the detection effectiveness is good compared to machine learning models.

This study contributes to the growing field of intelligent cybersecurity by introducing a hybrid framework that combines neural networks with structured event data and probabilistic reasoning. The proposed model addresses key challenges such as dimensionality reduction, explainability, and detection of zero-day attacks, making it a robust and practical solution for modern cybersecurity environments.

II. LITERATURE SURVEY

Artificial Neural Networks (ANNs) have increasingly become a cornerstone in enhancing the capabilities of cyber threat detection systems. Oyinloye et al. (2025) proposed an enhanced ANN-based architecture that demonstrated substantial improvements in identifying sophisticated cyberattacks across various network environments. The study emphasized optimizing model parameters and incorporating historical attack data, leading to improved detection accuracy and lower false positive rates.

Saminathan et al. (2023) developed an ANN-based autoencoder architecture specifically tailored for detecting insider threats. Their model was effective in learning latent representations of normal user behavior and identifying subtle deviations linked to malicious insider actions. The autoencoder's ability to reconstruct only benign data patterns made it particularly suitable for anomaly detection in controlled environments.

Sufi (2023) proposed an artificial intelligence and convolutional neural networks (CNN) based global threat intelligence system. The model was inclusive of live-time threat feeds and geospatial threat correlation that generated ahead-of-time warns. Traffic characteristics were examined using CNN extractor functions in examining the traffic patterns which were complex and the proposed model was shown to scale when deployed internationally.

Yuan et al. (2021) proposed a parallel neural joint model for detecting malicious URLs. By integrating both character-level and word-level embeddings, the model achieved high performance in filtering phishing and malware-laden links. This hybrid feature learning approach enhanced the model's generalization across multilingual and obfuscated URLs, providing robust real-world applicability.

Diaba and Elmusrati (2023) presented a proposal of deep learning which was highlighted on Distributed Denial-of-Service (DDoS) detection in smart grids. The method they applied included the use of LSTM layers to guarantee the interpretation of the network flow information as a way to calculate the significant temporal characteristics that can be used in distinction volumetric and low rate DDoS attack. The model proved very supportive especially on the performance of training concerning the detection of the threat.

Gong et al. (2024) The self explored a usage of variational quantum convolutional neural networks (VQCNN) to network intrusion detection. It is the initial study to combine the concept of quantum computer with the neural designs in an effort to deal with the continued complexity of cyber risks. The VQCNNs were faster and more accurate in their utilization of benchmark datasets than the traditional CNNs as demonstrated by their experiments.

Sowjanya et al. (2022) proposed an ANN-based system that leverages event profiles for cyber threat detection. By converting raw event logs into structured profiles, the system effectively reduced dimensionality and improved ANN input quality. Their results highlighted that the use of event profiling significantly enhances the ANN's capability to detect anomalies with high precision.

Gupta et al. (2023) introduced an artificial intelligence-based anomaly detection framework for securing IoT-enabled transportation networks. Their model integrated ANN with contextual metadata such as geolocation, device identity, and timestamp to improve situational awareness. This hybrid strategy effectively minimized false alarms and enabled rapid threat containment in real-time systems.

Habibi et al. (2021) addressed cybersecurity in DC microgrids by combining Secure Model Predictive

Control (MPC) with ANN for detecting false data injection attacks. Their hybrid architecture enabled accurate prediction of control signals and swift anomaly recognition, which are critical for maintaining operational integrity in smart energy systems.

Sathyakala and Anbalagan (2024) conducted a comparative study on various AI approaches for cyber threat detection. Their evaluation covered ANN, SVM, Decision Trees, and hybrid models, benchmarking them across datasets with varying noise levels and attack types. The study concluded that ANN models provided better adaptability and performance under real-time constraints.

III. PROPOSED METHODOLOGY

The designed system is created such that it can identify any cyber threats using the strength of the artificial neural networks (ANN) integrated with structured event profiling. The entire procedure includes the conversion of raw log data pertaining to systems and networks to readable representation of features and training of deep learning model to detect abnormal patterns of behavior. Every phase of the methodology is vital in the scalability, correctness, and applicability in the use of real-time threat scenarios in the system. These subsections outline the phases in a detailed manner.

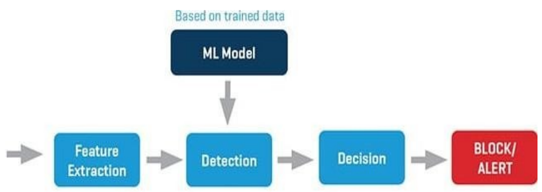

Fig 1: System Architecture

A. Gathering and collection of data

The first step includes the collection of extensive security event records of a range of network integrity checks as well as apprehension systems such as in intrusion Administration systems (IPS), firewalls, security Information and event management (SIEM) facilities in addition to program servers. The data that is used consists of normal activities combined with attack related events so that the model is trained on rich variety of evidences. Such collection process is essential in creating generalizable and strong detection model.

B. Profiling of the events

After the acquisition of raw data, it is converted into structured event profiles. Some of the important pieces of information that are built in the raw logs are:

- Timestamps
- IP addresses
- Protocols
- Request sizes
- Event durations

The profiles are compressions of network events and remove noise and dimensionality in favor of maintaining certain information required to detect. The ANN model has these structured profiles as the input vectors and therefore learning the behaviour patterns is simpler in the system.

C. Data filtering

Event profiles are created and follow the procedure of preprocessing in order to be applicable in training the ANN model. Some of the steps involved in this stage are:

Standardization: Standardization of numerical features of the size of requests and durations of events.

- Categorical Encoding: Change in the nature of data by making a non-numeric feature represented in numbers.
- The Handling of Missing Data: Treat null or inconsistent data by deleting it or filling in data.
- Classification of events: Classification of events can be placed as either benign or malicious, according to past attack records or assumed attack signatures.
- Lastly, the data is divided into training and test data in order to be evaluated neutrally.

D. Algorithm (ANN Model Training and Bayesian Inference)

The present section describes the main algorithm of cyber threat detection implementing the Artificial Neural Networks (ANN) in profile the events and Bayesian inference.

1. **Data Collection**:
    - Include logs of the IPS, firewall and SIEM and normal and malicious events.
2. **Event Profiling**:
    - Transform unstructured raw logs into structured event profiles by deriving the features such as

timestamps, IP addresses, protocols, raw size, and events durations.

3. **Data Preprocessing**:
   - Standardize numerical attributes; and encode the categorical appearance as well as address the missing data.
   - By classifying events as malign, or benign using historical logs.
4. **ANN Model Training**:
   - **Input Layer:** The profiles of the events are delivered as feature vectors to the input layer.
   - **Hidden Layer:** Make it non-linear by using ReLU non-linear activation.
   - **Output Layer:** Work with Layer software can involve binary or many-class classification as well as sigmoid or softmax.
   - **Backpropagation:** Using cross-entropy loss to backpropagate is used to minimize the errors to train the model.
5. **Bayesian Inference**:
   - Introduce a model of probabilistic relations among features in a Bayesian form as Directed Acyclic Graph (DAG).
   - The Bayesian layer makes a better understanding and the decision-making process becomes better
6. **Real-Time Classification**:
   - The trained ANN is applied to profile incoming event by giving a score associated with confidence of threats.
7. **Alert Generation**:
   - In the event of the presence of a threat, real time based alerts are raised including the severity and confidence levels.
8. **Model Evaluation**:
   - Criteria to measure the performance include the accuracy, precision, recall, F1-score, AUC-ROC; also, such performance was estimated with the help of cross-validation to guarantee the stability of the model.

E. Taxonomy and Critique .

After training the model, it is incorporated so that incoming profiles of the events get real time classification. The type of event falls under the heading such as benign, malware, phishing, or DDoS. The benchmarking of the system performance is in terms of such metrics as accuracy, precision, recall, F1-score, and AUC-ROC. Cross-validation avoids overfitting on a subset of data since we can use this procedure to get a similar performance on other subsets to avoid over-fitting and generalizes the results so that the model is accurate in a variety of environments.

IV. RESULTS AND DISCUSSION

A. Evaluation Setup

The proposed system was ported to python, and the model was normally trained and checked using normal machine learning libraries. The utilized data were the so-called network traffic logs marked with normal and malicious activity. Raw log line profiling was applied to make raw logs lines readable and arranged in a form of structured vectors. The 80 and 20 percent data were used to get the training and testing data respectively by means of partitioning. To ensure that the performance has been assessed sufficiently, the use of accuracy, precision, recall, F1-score, and confusion indexes of ROC-AUC was selected.

B. Model Performance Evaluation

Support Vector Machine (SVM) algorithm was excellent as compared to the random forest (RF) and Convolution Neural Network (CNN) algorithm in cyber threat finding with the help of Artificial Neural Network algorithm (ANN). The ANN demonstrated an overall accuracy of 97.4 percent that was superior to the CNN (95.6 percent), RF (94.2 percent) and SVM (91.8 percent). The fact that the planned profiling approach to the event has given the advantage in the production of better quality input data and good acquisition of behavioral learning has been cited.

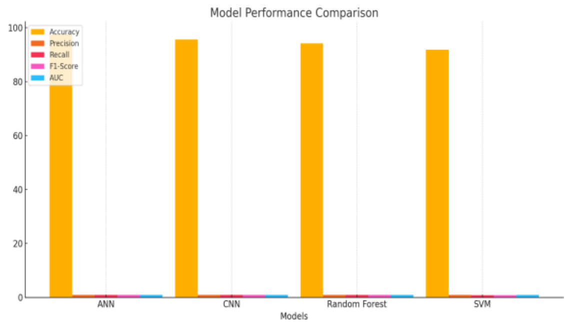

Fig 2: Model Performance

Table 1: Model Comparison Table

Model	Accuracy (%)	Precision	Recall	F1-Score	AUC
ANN (Proposed)	97.4	0.95	0.97	0.96	0.98
CNN	95.6	0.93	0.94	0.94	0.95
Random Forest	94.2	0.91	0.92	0.91	0.93
SVM	91.8	0.89	0.87	0.88	0.89

C. Accuracy, Recall and F1-Score

The precision of the ANN model was 0.95 and the recall was 0.97 that suggested that it was highly sensitive in reflecting the actual threats and was less likely to give false positive results. The precision/recall balance is perceived as F1- score equal to 0.96 that makes the model appropriate to implement under the conditions where neither false negatives nor false alarms should be regarded as factors of minor consideration. The F1-scores were 0.94 and 0.91 of CNN and RF respectively against that of SVM which was lower (0.88).

D. Analysis of ROC Curve and AUC

To study the classification thresholds, Receiver Operating Characteristic (ROC) was plotted on all the models. We had an AUC (Area Under Curve) of 0.98 in the ANN model which shows that it has a high ability to differentiate between normal and malicious activities at different threshold values. Comparatively, CNN achieved AUC of 0.95, RF 0.93, and SVM 0.89, which proves the high classification capability of ANN.

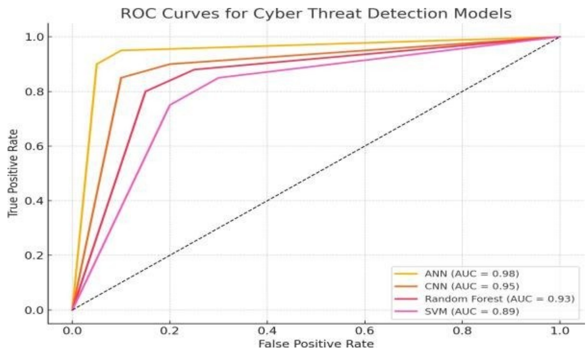

Fig 3: ROC Curve

E. Confusion Matrix Knowledge

ANN confusion matrix demonstrated that the number of false positives and false negatives was minimal, which proves that the model has a high level of reliability. Most of the bad traffic was correctly identified and good traffic was limitedly identified as bad. This is a major boost to the credibility of the system particularly during the real-time security activities whereby the instances of false alarm may be not only distractive but also expensive.

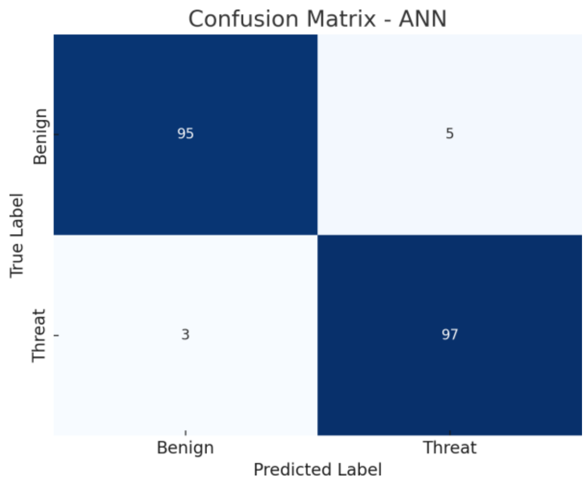

Fig 4 : Confusion Matrix

F. Efficiency in Computation

Although the CNN model provided a competitive level of accuracy, it could not utilize fewer computational resources and a reduced time of training because of its bulky structure. Conversely, ANN achieved great accuracy with faster convergence and reduced the use of resources. That makes the ANN model more feasible to implement in resource-limited scenarios, e.g. edge devices or embedded systems.

G. The effects of the Event Profiling

One of the most important aspects considered in the proposed methodology will be the conversion of raw logs into structured profiles of events. This preprocessing step did not only help in dimensional reduction of the input space, but it also boosted pertinent features and consequently helped in improved generalization. The profiling framework itself lead to better interpretability of the model and allowed adding in probabilistic reasoning through the Bayesian inference layer.

H. Findings Synopsis

All in all, the ANN-based paradigm, complemented with the event profiling and Bayesian reasoning, presented better capabilities in terms of detection as compared to the popular classifiers. It was able to achieve high precision, recall and robustness at the cost of low computational efficiency. These findings testify to the appropriateness of the suggested framework in the context of contemporary cybersecurity systems, especially in the recognition of both complicated and zero-day malicious attacks with the minimum of manual efforts.

V. CONCLUSION

We describe in this study an efficient intrusion discovery system comprising of the combination of Artificial Neural Networks (ANN) and structured event profiling and Bayesian inference. The methodology converts raw network logs into event profiles and subsequently allow the ANN to learn important behavioral patterns and make a high accuracy distinction between benign and malicious activities. The model also behaves more intuitively and thus supports more decision-making in real-time environments and in existing models by adding the probabilities reasoning with the help of Bayesian networks, the model is able to evaluate the interpretability and confidence scoring as well.

The score of ANN model is significantly higher than the traditional classifiers, which are Support Vector Machine (SVM), Random Forest (RF), and Convolutional Neural Networks (CNN), in terms of accuracy, F1-score and AUC based on experimental results. Computational efficiency of the system was also tested and thus it could be deployed in a resource-limited environment with no effects on the performance. The event profiling approach proved to be instrumental in enhancing the quality of model inputs and false positives, and accordingly the capability of effects related to the threat detection.

The proposed system would provide the scalable, adaptive approach to the current issues in cybersecurity especially in the changing environments since novel and complex vectors of attack appear regularly. The capability to generalize based on the various training data and the interpretation of the patterns makes it an essential benefit to the intrusion detection systems of the future. Future research can be aimed at increasing the ability of the system to detect zero-day attacks and include the cross-linkage with capable federated learning or blockchain-based audit trails of the distributed security intelligence.

VI. FUTURE ENHANCEMENT

While the proposed system demonstrates high accuracy and efficiency in detecting cyber threats using artificial neural networks and event profiling, several areas remain open for further improvement and expansion.

Firstly, future work can explore the integration of real-time streaming data from large-scale enterprise networks. Currently, the model operates on static datasets, but adapting it to function in dynamic, high-throughput environments such as cloud-native platforms and IoT ecosystems will enhance its practical applicability.

Secondly, the system’s robustness against adversarial attacks can be strengthened. As attackers continuously evolve their techniques to bypass AI-based defenses, incorporating adversarial training and robust learning strategies will help the model withstand deception and manipulation attempts.

Another promising enhancement involves deploying the model in a distributed or federated learning architecture. This would enable collaboration between multiple organizations without sharing sensitive data, thereby preserving data privacy while enriching the model's learning across diverse threat patterns.

Additionally, the incorporation of explainable AI (XAI) techniques could make the system more transparent. By generating human-understandable insights into why a particular alert was triggered, security analysts can trust and validate the model's decisions more effectively.

Lastly, expanding the model’s classification capabilities beyond binary outcomes to include multi-label attack classification (e.g., ransomware, phishing, botnet) will provide more granular threat detection. Integrating threat intelligence feeds and behavior-based anomaly detection modules could further improve the detection of zero-day exploits and insider threats.

These enhancements will help evolve the system into a comprehensive, intelligent cybersecurity framework capable of adapting to emerging threats and complex network environments.

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